FACTORY SIMULATION — PROGRESS REPORT

OOP with C++ GIST EECS

1. Team Information

Team name	BBANG
Factory domain	Smart Farm Bakery
GitHub repository	https://github.com/eh-rud/BBANG.git

Member	Student ID	Primary responsibility
김도경	20255021	Leader & Frontend Architecture & Interface Design (Developing Dear ImGui UI layouts, prototyping View layers, and implementing UI-to-Controller Command/Snapshot DTO bindings)
황서현	20255243	Backend Core Logic & State Machine (Designing the Factory Simulation Object hierarchy, implementing concrete machine process timers, and managing core update loops)

2. Class Inventory

List every class you have defined or plan to define. Mark its layer and whether it is abstract or concrete.

Class name	Layer (Model/View/Controller)	Abstract / Concrete	Inherits from	Status (Planned/In progress / Done)
SimulationObject	Model	Abstract		In progress
Source	Model	Abstract	SimulationObject	In progress
AbstractMachine	Model	Abstract	SimulationObject	In progress
Sink	Model	Abstract	SimulationObject	In progress
Worker	Model	Abstract	SimulationObject	In progress
StorageUnit	Model	Abstract	SimulationObject	In progress
Product	Model	Abstract		In progress
Order	Model	Abstract		In progress
SeedInput	Model	Concrete	Source	Planned
OrderGenerator	Model	Concrete	Source	Planned
SmartFarmGrower	Model	Concrete	AbstractMachine	Planned

Harvester	Model	Concrete	AbstractMachine	Planned
MillMachine	Model	Concrete	AbstractMachine	Planned
DoughMixer	Model	Concrete	AbstractMachine	Planned
Oven	Model	Concrete	AbstractMachine	Planned
PackagingMachine	Model	Concrete	AbstractMachine	Planned
ParcelSorter	Model	Concrete	AbstractMachine	Planned
OrderMatcher	Model	Concrete	AbstractMachine	Planned
DispatchStation	Model	Concrete	Sink	Planned
Technician	Model	Concrete	Worker	Planned
WheatSeed	Model	Concrete	Product	Planned
GrownWheat	Model	Concrete	Product	Planned
WheatBatch	Model	Concrete	Product	Planned
FlourBatch	Model	Concrete	Product	Planned
DoughBatch	Model	Concrete	Product	Planned
Bread	Model	Concrete	Product	Planned
BreadParcel	Model	Concrete	Product	Planned
SortedParcel	Model	Concrete	Product	Planned
DispatchedParcel	Model	Concrete	Product	Planned
BreadOrder	Model	Concrete	Order	Planned
WheatStorage	Model	Concrete	StorageUnit	Planned
FlourSilo	Model	Concrete	StorageUnit	Planned
DoughQueue	Model	Concrete	StorageUnit	Planned
BreadStorage	Model	Concrete	StorageUnit	Planned
ParcelQueue	Model	Concrete	StorageUnit	Planned
OrderQueue	Model	Concrete		Planned
EventLog	Model	Concrete		Planned
Factory	Controller / Model	Concrete		In progress
MainLoop	Controller	Concrete		Planned
UIManager	View	Concrete		Planned
SimulationCmd	Controller / View DTO	Struct		Planned
MachineCmd	Controller / View DTO	Struct		Planned
FactorySnap	Model / View DTO	Struct		Planned
MachineSnap	Model / View DTO	Struct		Planned
StorageSnap	Model / View DTO	Struct		Planned

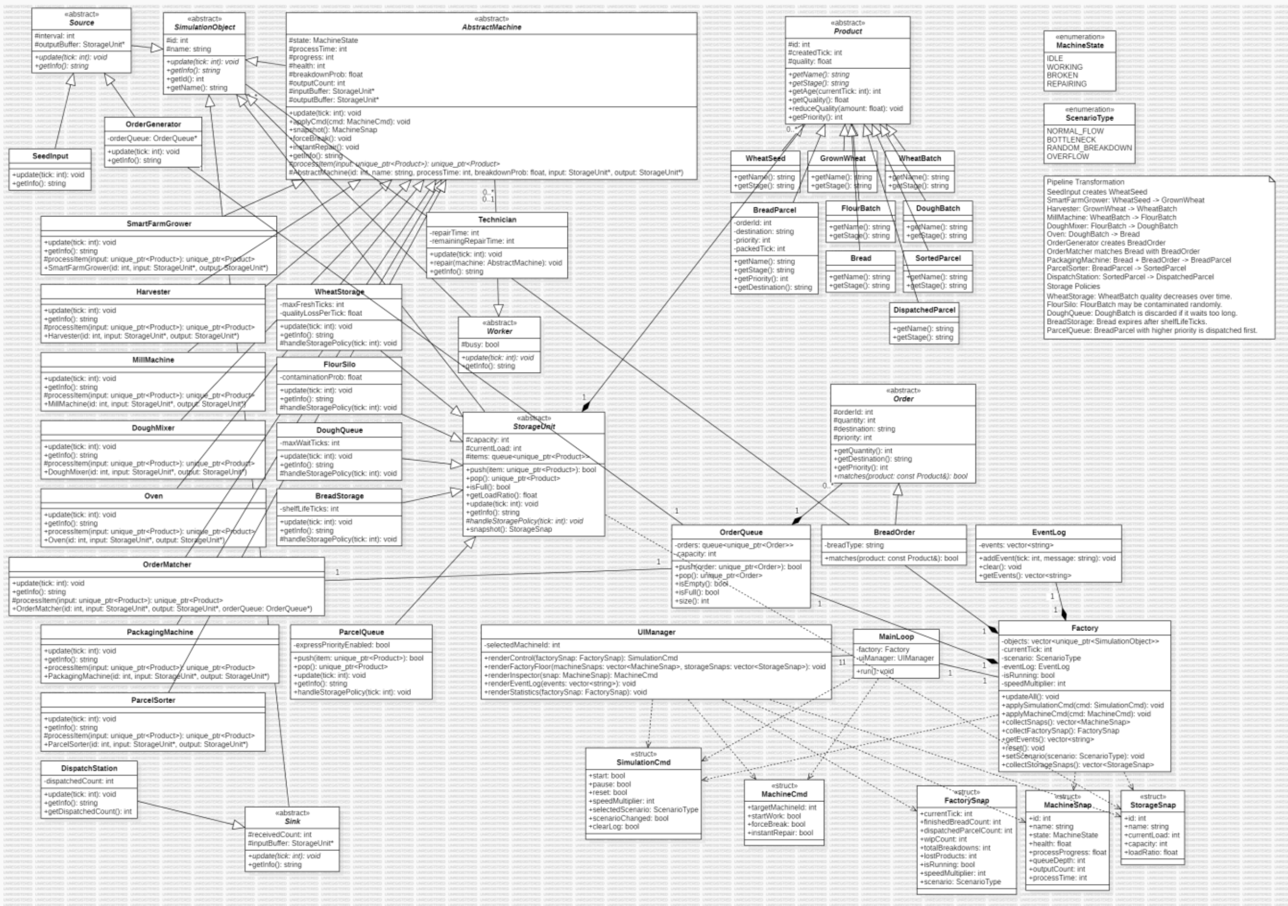
MachineState	Model	Enumeration		Planned
ScenarioType	Model	Enumeration		Planned
SimulationController	Controller	Concrete		In progress
MainWindow	View	Concrete		In progress
FactoryView	View	Concrete		In progress
ControlPanelView	View	Concrete		In progress
EventLogView	View	Concrete		In progress
InspectorView	View	Concrete		In progress
StatisticsView	View	Concrete		In progress
MachineWidget	View	Concrete		In progress
StorageWidget	View	Concrete		In progress

Layer: Model/View/Controller Status: Planned / In progress / Done

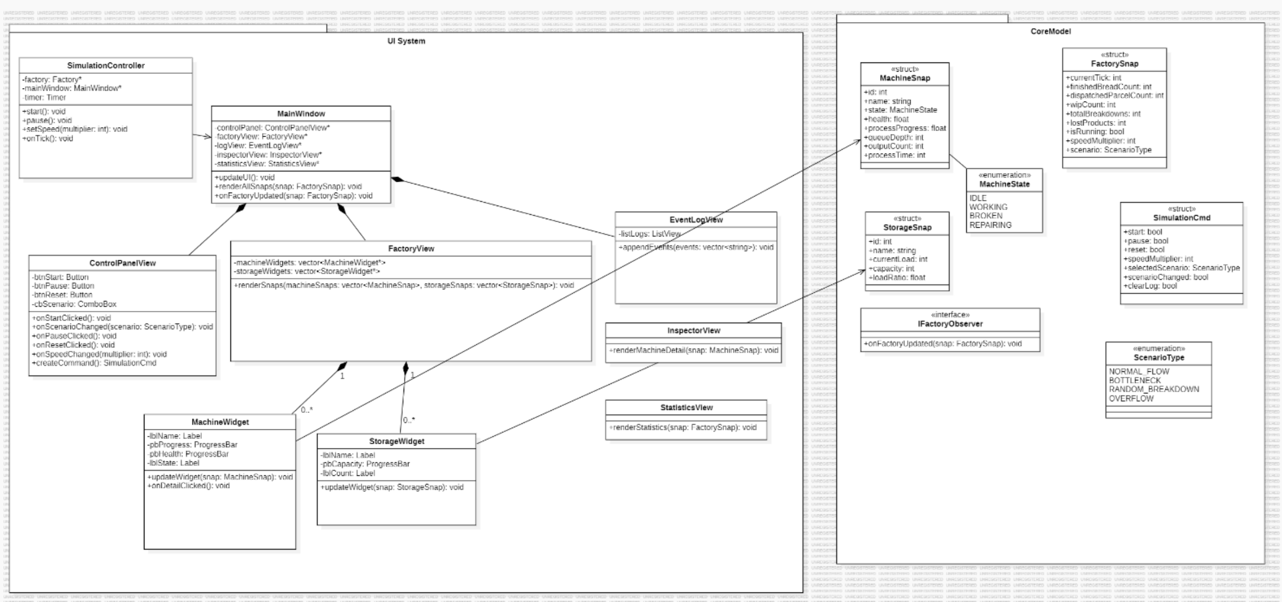
3. Class Diagram (first version)

Attach or paste your current UML class diagram below. Hand-drawn is fine at this stage. Must show: class names, inheritance arrows, and key methods.

bread (3).mdj UI_final.mdj



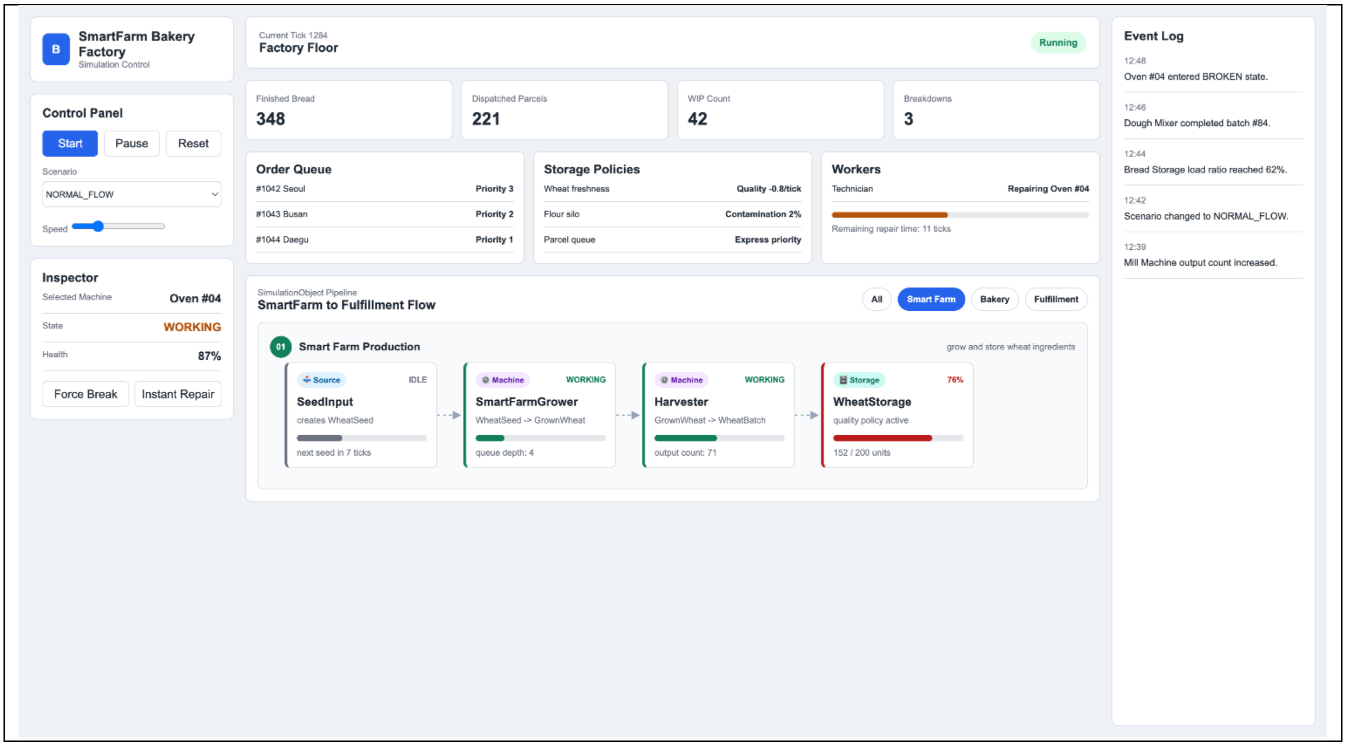
1. Code UML



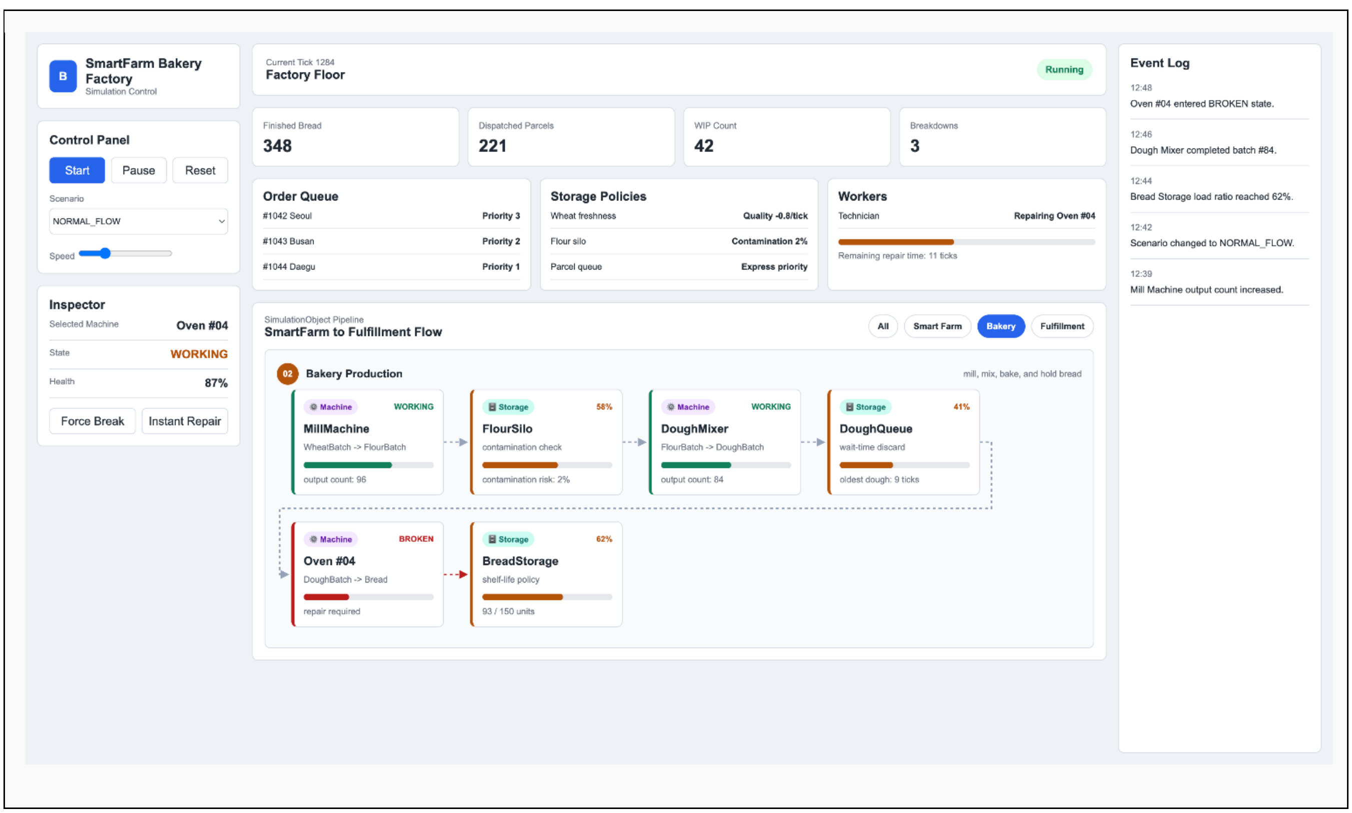
2. UI UML

4. UI Screenshots

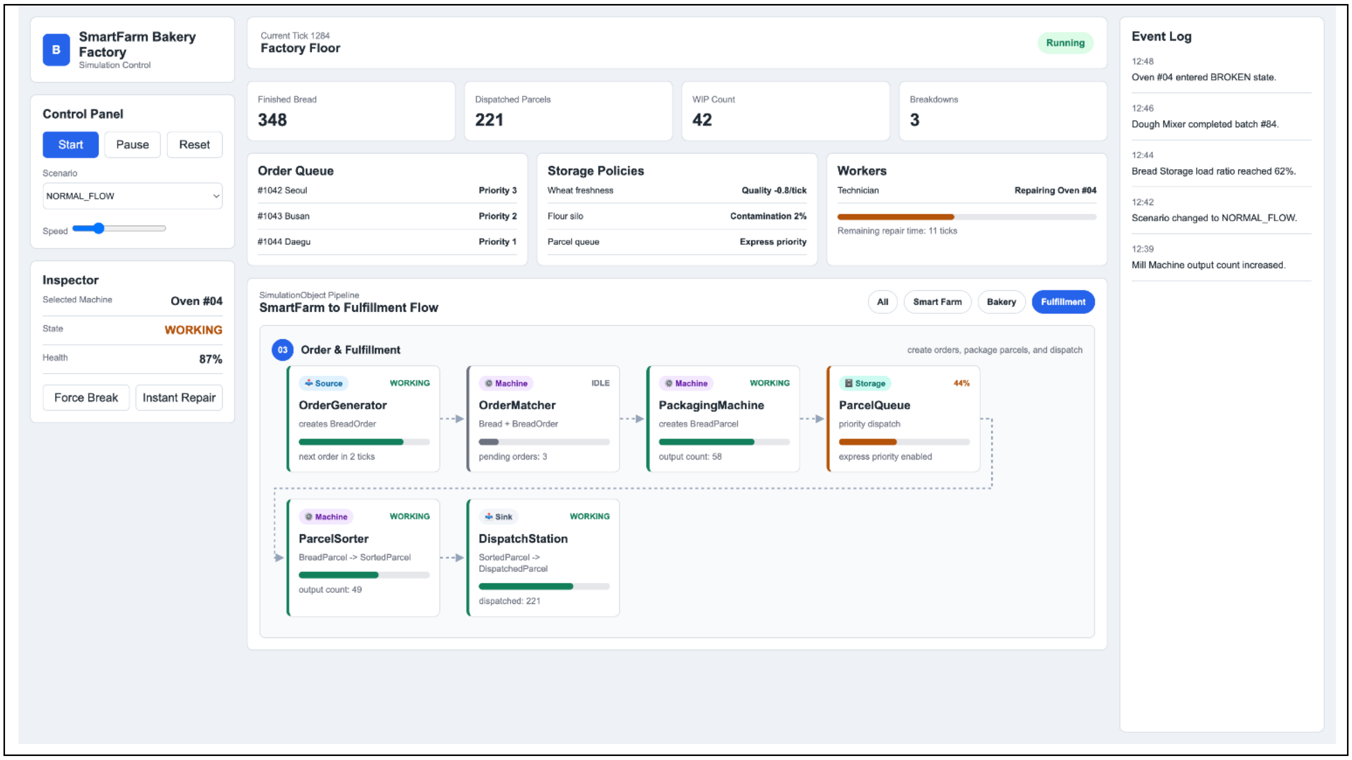
Paste 1–3 screenshots of your current UI. Add a one-line caption below each one explaining what it shows.



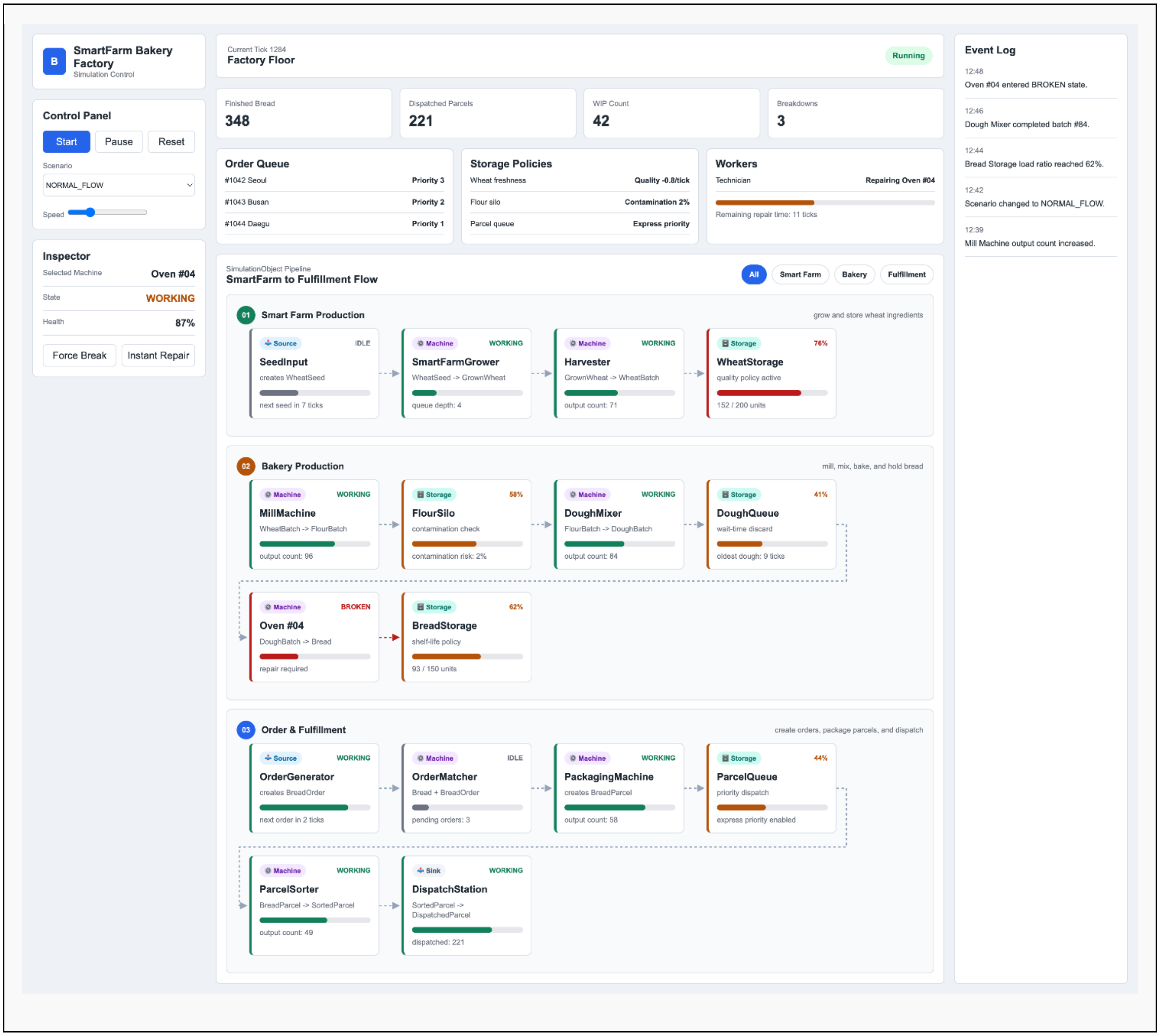
Caption: Smart Farm production view, monitoring the initial agricultural process from seed input to wheat storage.



Caption: Bakery production view, illustrating color-coded machine states such as the BROKEN status of the oven.



Caption: Order and fulfillment view, detailing order matching, packaging, and final dispatch metrics.



Caption: Main dashboard overview displaying the complete Smart Farm Bakery pipeline, control panel, and real-time event log.

5. Project Completion Roadmap

List your remaining tasks with target dates. Be specific — one task per row.

#	Task	Owner	Target date	Done?
1	Complete the core ImGui UI layout and component prototyping (Smart Farm View, Inspector, Event Log, Statistics) without backend linkage	김도경	May 24	■
2	Define SimulationCmd and MachineCmd structures and implement event bindings for UI-to-Controller command delivery	김도경	May 26	□

3	Connect the initial pipeline (SeedInput ➔ SmartFarmGrower ➔ Harvester ➔ WheatStorage) and verify the pure model update(tick) loop	황서현	May 28	◻
4	Implement core state machine logic and processing timers for late-stage bakery machines (MillMachine, DoughMixer, Oven, PackagingMachine, etc.)	황서현	May 30	◻
5	Implement snapshot structures (MachineSnap, StorageSnap) to extract and capture real-time backend state data for the UI layer	김도경	June 1	◻
6	Map scenario selection logic and implement backend policy branching for the 4 major scenarios (NORMAL_FLOW, RANDOM_BREAKDOWN, etc.)	황서현	June 3	◻
7	Conduct full system integration testing (debugging real-time UI data updates, checking for memory leaks) and configure GitHub Pages	Joint	June 5	◻
8	Prepare the final presentation materials, record the simulation demo video, finalize the progress/final report, and push to GitHub	Joint	June 6	◻

6. Instructor Feedback

(This section is for instructor use only.)

Progress rating	◻ On track ◻ Minor delays ◻ At risk ◻ Needs discussion
Architecture check	◻ Frontend / backend correctly separated ◻ Needs revision
UML quality	◻ Satisfactory ◻ Incomplete ◻ Incorrect — resubmit
Comments	
Date reviewed	